

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Civil) (Credit System) Degree Examinations**

May - June 2023

20CV601 – DESIGN OF STEEL STRUCTURAL ELEMENTS

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Use of IS 800: 2007, Steel tables is permitted.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) List out advantages and disadvantages of steel structures over RCC structures.	10	L*2	1	1,2
	b) Determine the bolt value for butt joint using M20 property class 5.6 bolts with following data, two cover plates of 6mm thick, main plate of 10mm thick.	10	L2	2	1,2,3
2.	a) Evaluate ultimate tensile stress and yield stress in property class 4.6, 5.6, and 8.8 grade bolts.	10	L2	1	1,2,3
	b) With neat diagrams discuss various modes of failure in bolted connections.	10	L2	2	1,2,3
3.	a) Discuss various modes of failures in tension member	10	L2	1	1,2
	b) Design a tie member consisting of single angle section to carry a working load of 150kN use bolted connection with M18 property class 5.6 bolts, if length of the member is 2m check for slenderness ratio.	10	L2	2	1,2,3
Unit – II					
4.	a) Define buckling? Discuss Local Buckling, Flexural buckling and Torsional- flexural buckling.	10	L2	3	1,2
	b) Design the compressive strength of a rolled steel angle section ISA 100 X 100 X 8 mm @ 12.1 Kg/m with the length of the member is 2.5m. Assume $f_y = 250$ mPa and the ends are hinged.	10	L2	4	1,2,3
5.	Design a compression member using double channel section face to face to carry a factored load of 1600kN, the length of the column is 5m, with one end fixed and one end hinged.	20	L2	3	1,2,3
6.	Design a slab base for a column ISHB 300@58.8kg/m subjected to a service load of 1500kN. The grade of concrete for pedestal is M20 and SBC of soil is 180kN/m ² . Design slab base and concrete base with welded connection.	20	L2	3	1,2,3
Unit – III					
7.	a) Discuss detailed design procedure for laterally supported beams as per IS 800:2007.	10	L2	5	1,2
	b) Define laterally supported and laterally unsupported beams.	10	L2	5	1,2
8.	Design a Seating angle connection for an ISMB 400 beam to an ISHB 200 column so as to transfer a shear of 150 kN.	20	L2	5	1,2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
